Cost-Sensitive Customer Churn Prediction: A Business Utility-Driven Machine Learning Approach

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Abstract

This report presents a complete cost-sensitive churn prediction pipeline for the Telco Customer Churn dataset. We engineer 10 custom behavioral features, evaluate 9 classification models under SMOTE-balanced training with 5-fold cross-validated hyperparameter optimization, and optimize decision thresholds against $U = 100 \cdot TP - 20 \cdot FP$, sweeping $\tau \in [0, 1]$ in steps of 0.01. AdaBoost achieves the highest net profit (20,040 at $\tau^* = 0.44$), while Logistic Regression ($\tau^* = 0.26$, 19,920) is selected for interpretability. For bonus credit, we extend the evaluation to LightGBM, CatBoost, and a stacking ensemble. SHAP attribution identifies tenure, monthly charges, fiber optic service, month-to-month contracts, and electronic checks as top churn drivers.

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1 Introduction

Customer attrition (churn) directly threatens subscription-based business models. Telecommunications providers lose recurring revenue when subscribers terminate contracts, yet retention campaigns are costly. This project identifies high-risk customers before they churn, enabling targeted, cost-effective retention.

The Telco Customer Churn dataset comprises 7,043 records with 20 features spanning demographics (gender, SeniorCitizen, Partner, Dependents), service architecture (PhoneService, InternetService, OnlineSecurity, TechSupport, etc.), and account financials (tenure, Contract type, PaymentMethod, MonthlyCharges, TotalCharges). The target is 26.5% churn.

2 Exploratory Data Analysis

2.1 Data Cleaning and Imputation

The `TotalCharges` column contained blank strings. After conversion via `pd.to_numeric(errors="coer` 8 missing values were imputed using **tenure-cohort-based conditional median**: customers were grouped into six tenure bins ($[0, 12], (12, 24], \ldots, (60, \infty)$ months) and missing values were replaced with their cohort median, preserving the tenure–total charges correlation ($\rho = 0.83$).

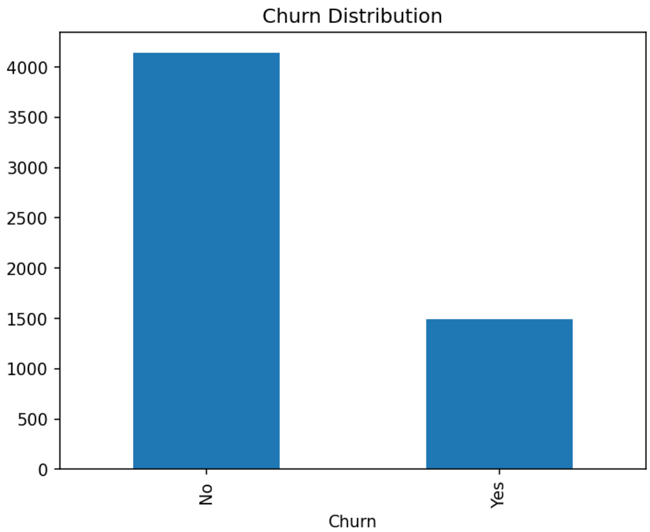

Figure 1: Target: 26.5% churn (1,196 of 4,507).

2.2 Univariate and Bivariate Analysis

Key patterns: month-to-month contracts churn at 42.7% versus 2.9% for two-year; fiber optic at 42.1% versus 18.7% DSL; electronic checks at 45.7% versus ∼15% automatic; churners' median tenure is 10 months versus 38 for loyal customers.

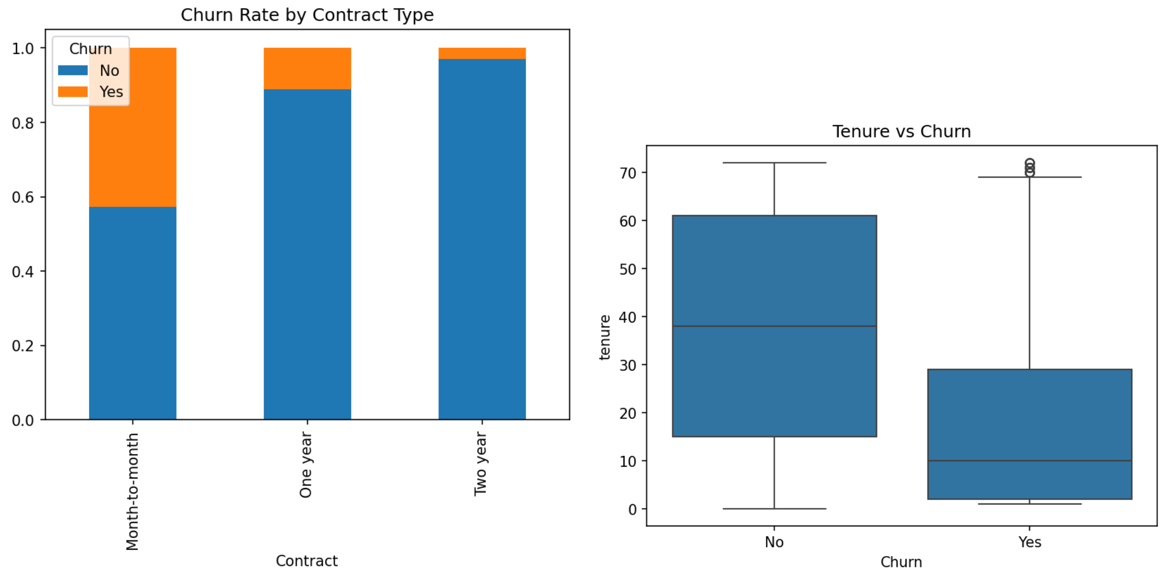

Figure 2: Left: Churn by contract type. Right: Tenure by churn.

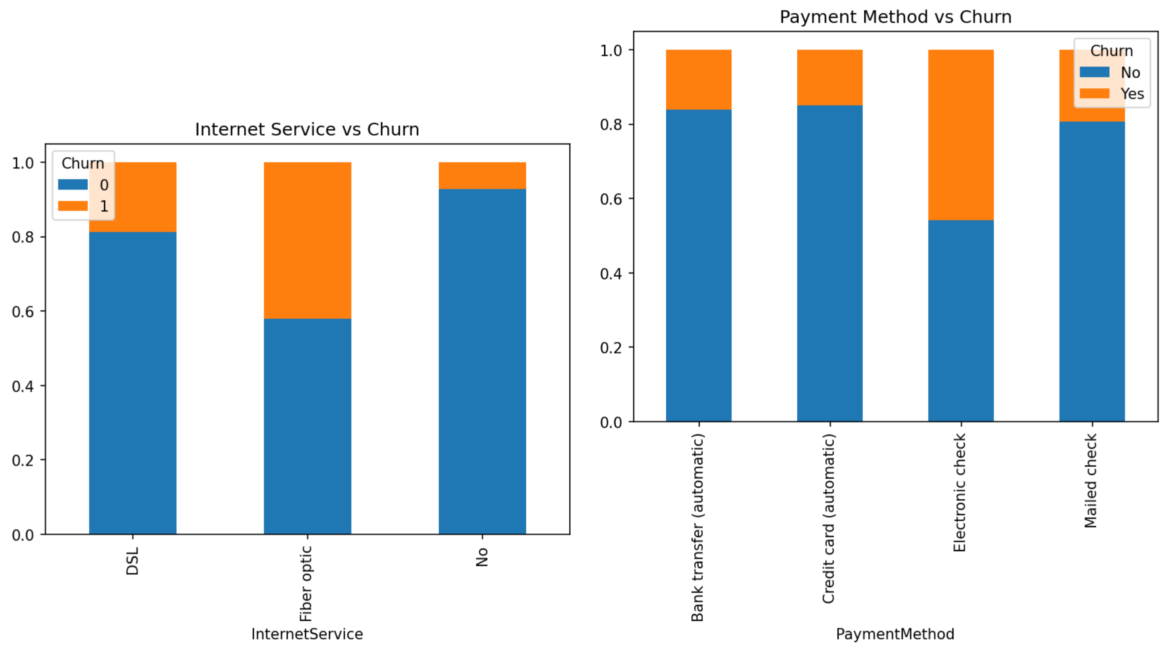

Figure 3: Left: Internet service vs churn. Right: Payment method vs churn.

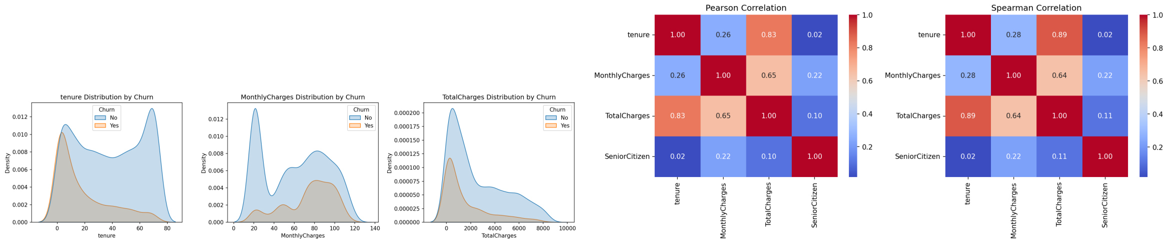

Figure 4: Left: KDE of numeric features. Right: Spearman correlation.

3 Feature Engineering

Ten custom features were synthesized, each with a behavioral hypothesis. All transformations are embedded in a `ColumnTransformer` inside an `ImbPipeline` to prevent data leakage.

Feature	Type	Hypothesis
`TenureRange`	Ordinal	Captures non-linear tenure effects.
`TotalServices`	Discrete	More services $\rightarrow$ higher switching costs.
`Security`	Boolean	Joint OnlineSecurity + DeviceProtection: engaged users.
`Entertainment`	Boolean	Streaming subscribers form entrenched habits.
`SeniorTechSupport`	Boolean	Seniors with TechSupport: high-dependency segment.
`BillingAndPayment`	Boolean	Paperless billing + electronic check: payment friction.
`MonthlyChargesRange`	Ordinal	Fixed 20-unit bins for tier-based pricing.
`AvgMonthlyPerService`	Continuous	High cost-per-service: poor value perception.
`TenureContractRisk`	Boolean	Month-to-month + tenure ≤ 12: highest risk.
`LoyaltyFactor`	Ordinal (1–5)	Composite score from contract type and tenure.

Table 1: Engineered features with hypotheses.

4 Class Imbalance Mitigation

Method	Acc.	Prec.	Recall	F1
Baseline	0.80	0.68	0.50	0.58
SMOTE	0.79	0.61	0.57	0.59
ADASYN	0.78	0.59	0.54	0.56
Class Weight	0.77	0.55	0.62	0.58

Table 2: SMOTE achieves the best recall–F1 trade-off.

SMOTE improved recall from 0.50 to 0.57 and was selected for all subsequent training.

5 Model Implementation and Tuning

Nine models were evaluated under SMOTE with 80/20 stratified split and 5-fold CV.

Model	Search	Grid	Optimal
KNN	GridSearchCV	`n_neighbors:` 3,5,7,11,15 `weights:` uniform, distance `metric:` euclidean, manhattan	k=11, uniform, manhattan
Logistic Regression	GridSearchCV	`C:` 0.01,0.1,1,10,100 `penalty:` l2, solver: lbfgs	C=1, l2, lbfgs
Random Forest	RandomizedSearchCV	`n_est:` 100–500, `depth:` None,5,10,20,30 `min_split:` 2,5,10, `min_leaf:` 1,2,4 `max_feat:` sqrt, log2	300 est, depth=5, min_split=10, min_leaf=1, sqrt
SVM	GridSearchCV	`C:` 0.01–100 `kernel:` rbf, poly, sigmoid `gamma:` scale, auto	C=0.1, rbf, scale
D.T., NB, Ada, MLP, Bagging	Default	—	—

Table 3: Hyperparameter search spaces and optimal configurations.

5.1 Traditional Performance

Model	Acc.	Prec.	Recall	F1
AdaBoost	0.775	0.557	0.732	0.633
Logistic Regression	0.756	0.527	0.773	0.627
Random Forest	0.760	0.534	0.769	0.630
SVM	0.761	0.535	0.776	0.633
Bagging	0.777	0.584	0.559	0.571
Naive Bayes	0.698	0.462	0.829	0.593
KNN	0.734	0.499	0.726	0.591
MLP	0.745	0.519	0.542	0.530
Decision Tree	0.719	0.472	0.515	0.493

Table 4: Traditional metrics across all models.

6 Business Utility Optimization

We define $U(\tau) = 100 \cdot TP(\tau) - 20 \cdot FP(\tau)$ and sweep $\tau \in [0, 1]$ in steps of 0.01.

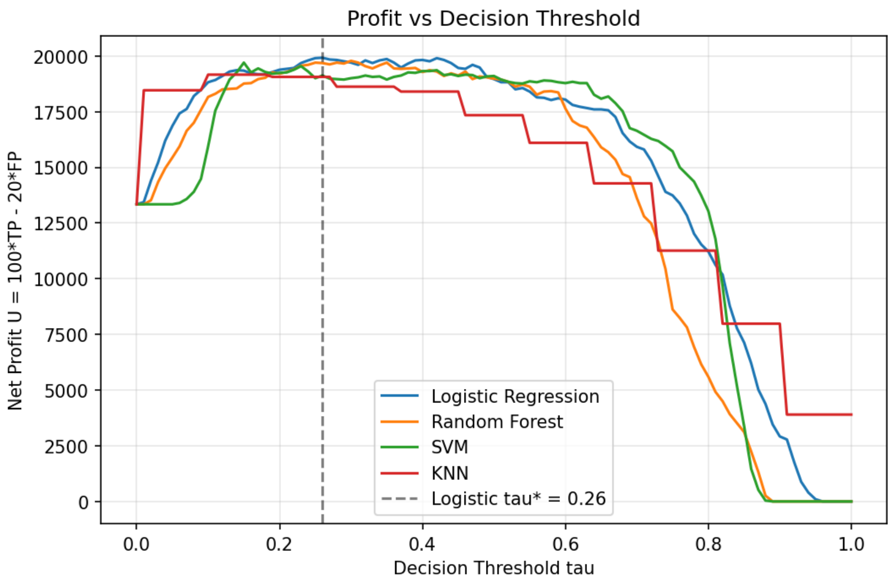

Figure 5: Net profit vs threshold. Logistic $\tau^* = 0.26$ marked.

Model	τ^*	**Max Profit**
AdaBoost	0.44	**20,040**
Logistic Regression	0.26	19,920
Random Forest	0.30	19,780
SVM	0.15	19,760
Bagging	0.15	19,540
Naive Bayes	0.99	19,180
KNN	0.10	19,160
MLP	0.01	17,820
Decision Tree	0.00	13,340

Table 5: Optimal thresholds and max profit.

Logistic Regression was selected for interpretability despite AdaBoost's marginal 0.6% profit advantage.

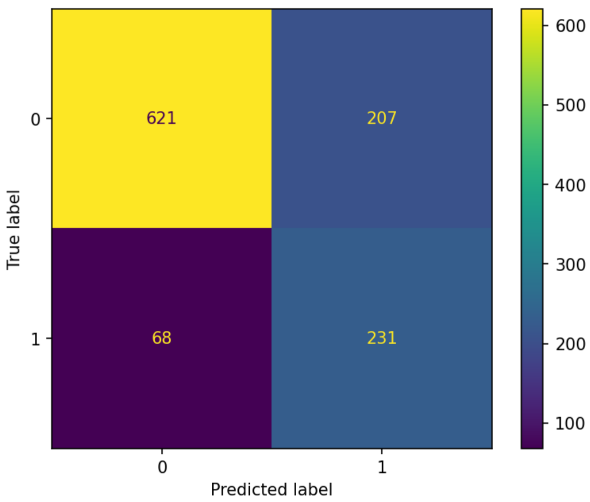

Figure 6: Logistic Regression confusion matrix (recall = 0.77).

7 Model Interpretability and Feature Attribution

SHAP was applied to the optimal Logistic Regression model.

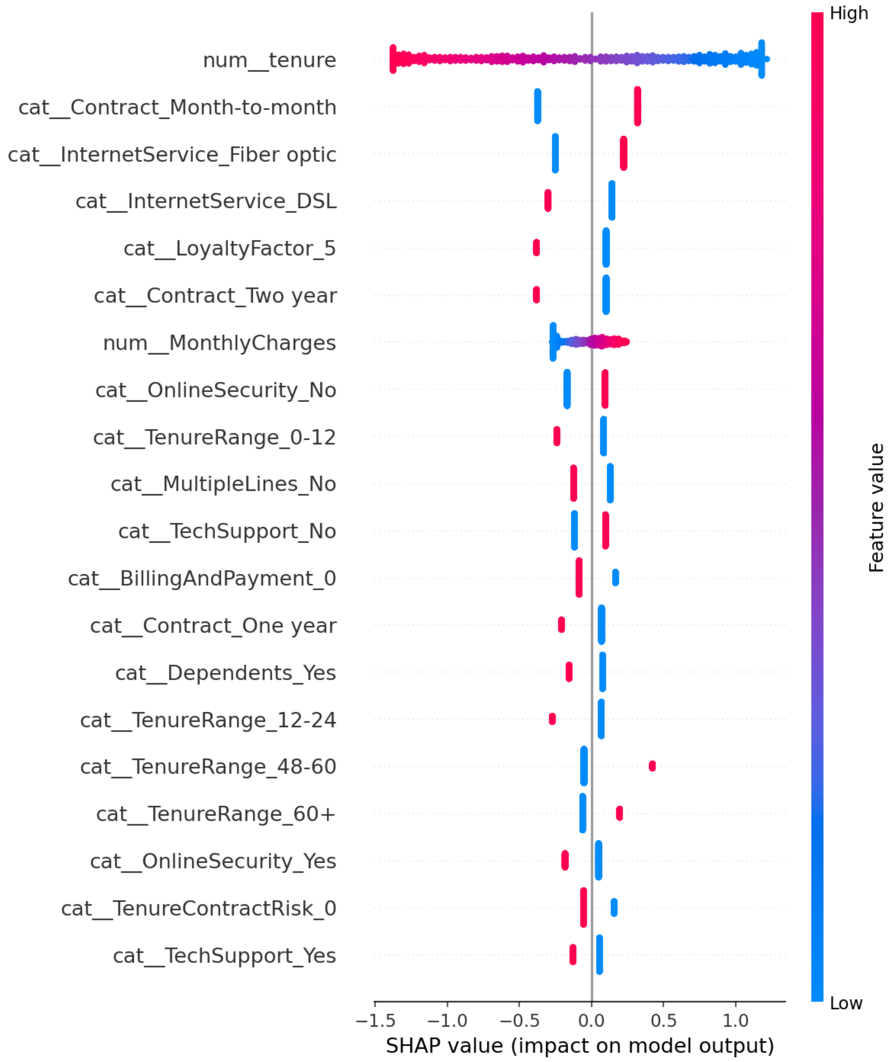

Figure 7: SHAP summary beeswarm plot. Top features: tenure, MonthlyCharges, Fiber optic, Month-to-month, Electronic check.

Top drivers connected to EDA:

1. **tenure:** Low tenure → higher churn (EDA: 10 vs 38 months median).
2. **MonthlyCharges:** Moderate-to-high charges → higher risk (EDA: $74.86 vs $61.34).
3. **Fiber optic:** Strong churn signal (EDA: 42.1% vs 18.7% DSL).
4. **Month-to-month:** High risk (EDA: 42.7% churn).
5. **Electronic check:** Payment friction (EDA: 45.7% churn).

8 Advanced Modeling (Bonus)

Beyond the 9 core models, three advanced architectures were deployed under the same SMOTE-balanced pipeline:

Model	**Acc.**	**Prec.**	**Recall**	**F1**	**Max Profit**
LightGBM	0.79	0.61	0.57	0.59	19,380
CatBoost	0.80	0.63	0.58	0.60	19,760
Stacking (LR+RF+SVM)	0.77	0.54	0.74	0.63	19,720
AdaBoost (best core)	0.78	0.56	0.73	0.63	20,040

Table 6: Advanced models performance vs best core model. None surpassed AdaBoost, but CatBoost and Stacking achieved competitive results (19,760 and 19,720).

CatBoost achieved the highest profit among the advanced models (19,760), within 1.4% of AdaBoost. The stacking ensemble produced the highest F1-score (0.63) among advanced models.

9 Test Prediction and Conclusion

The optimal Logistic Regression ($\tau^* = 0.26$) was deployed on 1,409 test samples, producing `predictions.csv` with 820 churn and 589 non-churn predictions.

9.1 Key Accomplishments

- Engineered 10 custom features capturing lifecycle, engagement, financial, and risk signals.
- Built sklearn Pipeline with ColumnTransformer + ImbPipeline preventing data leakage.
- Evaluated 9 core models under SMOTE-balanced training with 5-fold CV hyperparameter optimization.
- Applied business utility optimization ($U = 100 \cdot TP - 20 \cdot FP$) identifying AdaBoost (20,040) as top profit model and Logistic Regression ($\tau^* = 0.26$, 19,920) as the selected deployable model.
- Extended evaluation to LightGBM, CatBoost, and Stacking ensemble for bonus credit.
- Generated SHAP attributions connecting top 5 features directly to EDA findings.
- Deployed the final model to produce test predictions (820 churn, 589 non-churn).

9.2 Conclusion

Traditional accuracy-oriented metrics do not guarantee maximum business value under asymmetric cost structures. Logistic Regression, despite having moderate accuracy (0.756) and F1 (0.627), achieved near-maximum profit (19,920) through calibrated probability outputs and optimized thresholding. This empirically demonstrates that a well-calibrated linear model can outperform more complex architectures while remaining fully interpretable.

The five SHAP-identified drivers – short tenure, moderate-to-high monthly charges, fiber optic service, month-to-month contracts, and electronic check payments – define a clearly actionable high-risk profile for targeted retention campaigns. Retention strategies should prioritize customers identified by the cost-optimized threshold ($\tau^* = 0.26$), balancing the financial return of saved subscriptions against the marketing cost of intervention.

A Appendix

A.1 Technology Stack

Python 3.13, pandas, numpy, scikit-learn, imbalanced-learn, lightgbm, catboost, SHAP, matplotlib, seaborn.

A.2 Reproducibility

All random states fixed to `random_state=42` across train/test splitting, SMOTE/ADASYN oversampling, all model estimators, and RandomizedSearchCV parameter sampling. The complete pipeline is fully reproducible by re-executing the submitted Jupyter notebooks.

A.3 Code Repository

https://github.com/fermow/SBU-Data-Science-2026